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MT6686 Datasheet

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Version History

Version	Date	Description
0.1	2022-01-24	Initial draft
0.2	2022-07-20	- Corrected L1 and L5 channel label in Figure 1-1. - Updated VCN15 to VCN18 in Table 3-4. - Updated supply voltage information in Figure 3-1.
0.3	2022-08-17	- Updated Table 3-4. - Added note in Section 3.7.1.1 that MT2737 does not support E5b mode.
0.4	2022-09-02	Updated top marking format.
1.0	2024-01-30	- Document structure rearrangement - Added Chapter 2 Terms and Abbreviations. - Revised sections 1.1.2.1, 4.3, and 4.5. - Revised Table 4 7 and Table 4 9.
1.1	2024-07-24	- Corrected “50ms” to “50μs” in Section 4.3 Power On/Off Sequence. - Revised the ordering number in Table 7-1. MT6686 ordering information. - Added T_{stg} information in Table 4-2. Recommended operating range. - Added Section 7.4 Storage.
1.2	2024-08-12	Added Chapter 5 Part Reliability and Chapter 6 Packing Information (“Storage” Section is moved to 6.2 Storage)
1.3	2024-09-23	Added the maximum input level parameter to Table 4-9 to Table 4-13
1.4	2025-05-07	Updated the min. value of voltage swing and the typ. values of phase noise in Table 4-7. Reference clock requirement

Table of Contents

Version History	2
Table of Contents	3
List of Figures	5
List of Tables	5
1 Introduction	6
1.1 MT6686 Product Overview	6
1.1.1 Functional Block Diagram	6
1.1.2 Highlighted Features	7
2 Terms and Abbreviations	8
2.1 Naming Convention	8
2.2 Abbreviations	9
3 Pin Information	10
3.1 Pin Map	10
3.2 Pin Descriptions	11
4 Specifications	13
4.1 Electrical Characteristics	13
4.1.1 Absolute Maximum Ratings	13
4.1.2 Operating Descriptions	13
4.2 Supply Specifications	14
4.3 Power On/Off Sequence	15
4.4 Digital Logic Characteristics	16
4.4.1 Timing Diagram Convention	16
4.4.2 Rising/Falling Time Definition	16
4.4.3 Protocol	17
4.5 Reference Clock	18
4.6 GNSS	19
4.6.1 GNSS Radio Description	19
4.6.2 GNSS Mode Definition	19
4.6.3 GNSS Specifications	19
5 Part Reliability	22
5.1 Qualification Summary	22
5.2 Device Characteristics	23
6 Packing Information	24
6.1 Carrier	24
6.1.1 Tape and Reel Information	24
6.1.2 Aluminum Foil Bag	27
6.1.3 Box and Carton	27
6.2 Storage	27
6.2.1 Bagged	27
6.2.2 Out-of-Bag	27
7 Package Information	28
7.1 Top Marking	28
7.2 Ordering Information	28

7.3 Mechanical Drawing.....	29
Exhibit 1 Terms and Conditions.....	30

List of Figures

Figure 1-1. MT6686 block diagram	6
Figure 3-1. Overview of MT6686 pin map	10
Figure 4-1. Power-on sequence	15
Figure 4-2. Timing diagram conventions	16
Figure 4-3. Rising and falling times diagram	16
Figure 4-4. 2-wire SPI timing diagram	17
Figure 6-1. Carrier tape dimensions	24
Figure 6-2. Cover tape	25
Figure 6-3. Reel dimensions	26
Figure 7-1. Top marking of MT6686	28
Figure 7-2. Physical dimension of MT6686	29

List of Tables

Table 2-1. I/O definitions	8
Table 2-2. Abbreviations	9
Table 3-1. MT6686 pin descriptions	11
Table 4-1. Absolute maximum ratings	13
Table 4-2. Recommended operating range	13
Table 4-3. DVDD13_CEXT specifications	14
Table 4-4. AVDD_GNSS/AVDD_XO/DVDD_TOP specifications	14
Table 4-5. DVDD18_12_IO specifications	14
Table 4-6. Operating conditions of digital logics	17
Table 4-7. Reference clock requirement	18
Table 4-8. Pin (XO_IN) input impedance	18
Table 4-9. GNSS RF specifications (GPS-only mode or Galileo-only mode or GPS/Galileo mode)	19
Table 4-10. GNSS RF specifications (GPS/GLONASS mode or Galileo/GLONASS mode)	20
Table 4-11. GNSS RF specifications (GPS/BeiDou mode)	20
Table 4-12. GNSS RF specifications (GPS/GLONASS/BeiDou mode or GPS/GLONASS/Galileo mode)	21
Table 4-13. GNSS RF specifications (GPS-L5/Galileo E5a/IRNSS/BeiDou-3 B2a mode)	21
Table 5-1. MT6686 product reliability qualification	22
Table 5-2. MT6686 package reliability qualification	22
Table 5-3. MT6686 device characteristics	23
Table 6-1. Aluminum foil bag specification	27
Table 7-1. MT6686 ordering information	28

1 Introduction

1.1 MT6686 Product Overview

MT6686 is a GNSS L1-band and L5-band receiver front-end chip in a WLCSP package. It is a companion chip to 5G modems such as MT6880/90, MT6980/90 or MT2737, which come with the ADC (analog-to-digital converter) to process the GNSS receiver front-end signals. The interface driver/receiver buffer is designed to drive PCB trace loading. MT6686 does not have its own dedicated crystal oscillator. It uses either an external (temperature compensated) oscillator or clock source from the companion chips in the platform.

A simplified block diagram of MT6686 is shown in [Figure 1-1](#).

1.1.1 Functional Block Diagram

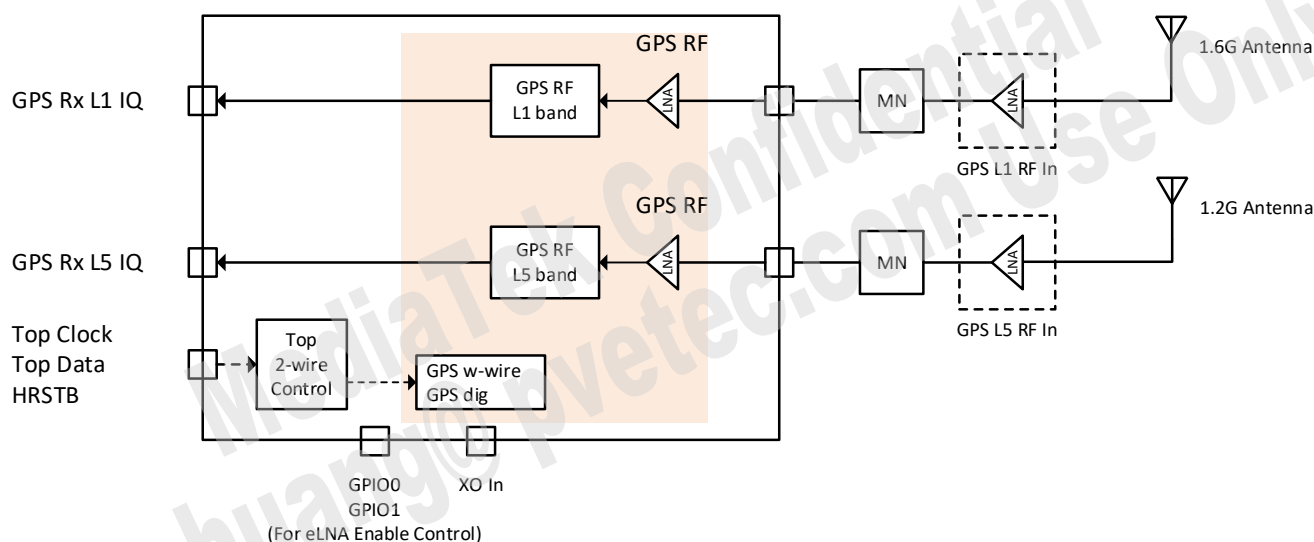


Figure 1-1. MT6686 block diagram

1.1.2 Highlighted Features

- MT6686 is a GNSS chip which contains a L1-band receiver front-end and a L5-band receiver front-end in a WLCSP package.
- MT6686 supports GNSS external LNA which can be controlled by the GPIO0 and GPIO1 pins via SPI interface with modem chips.

1.1.2.1 GNSS

- RF supports simultaneous dual-band L1 and L5 operation.
- RF supports GPS, Galileo, GLONASS and BeiDou.
- Simultaneous reception of GPS + GLONASS + BeiDou + Galileo for more accurate positioning
- Built-in calibrations for PVT variation
- Supports external LNA.
- Multi-mode filters for different GNSS receiver modes

2 Terms and Abbreviations

2.1 Naming Convention

Table 2-1. I/O definitions

Pad Attribute	
AI	Analog input (excluding pad circuitry)
AIO	Analog bidirectional (excluding pad circuitry)
AO	Analog output (excluding pad circuitry)
DI	Digital input (CMOS)
DIO	Bidirectional digital with CMOS input
GND	Ground
P	Voltage supply
High-Z	High-impedance output

2.2 Abbreviations

Table 2-2. Abbreviations

Abbreviation	Description
CMOS	Complementary Metal-Oxide-Semiconductor
DC	Direct Current
DIO	Digital Input/Output
ESD	Electrical Static Discharge
GNSS	Global Navigation Satellite System
GPIO	General Purpose Inputs-Outputs
GPS	Global Positioning System
I/O	Input/Output
IC	Integrated Circuit
IF	Intermediate Frequency
IQ	In-phase and Quadrature-phase signals
eLNA	(External) Low Noise Amplifier
LPF	Low Pass Filter
MHz	Megahertz
PCB	Printed Circuit Board
PGA	Programmable Gain Amplifier
PVT	Process, Supply Voltage and Temperature
RF	Radio Frequency
RX	Receiver
SAW	Surface Acoustic Wave
SPI	Serial Peripheral Interface
WLCSP	Wafer Level Chip Scale Package

3 Pin Information

3.1 Pin Map

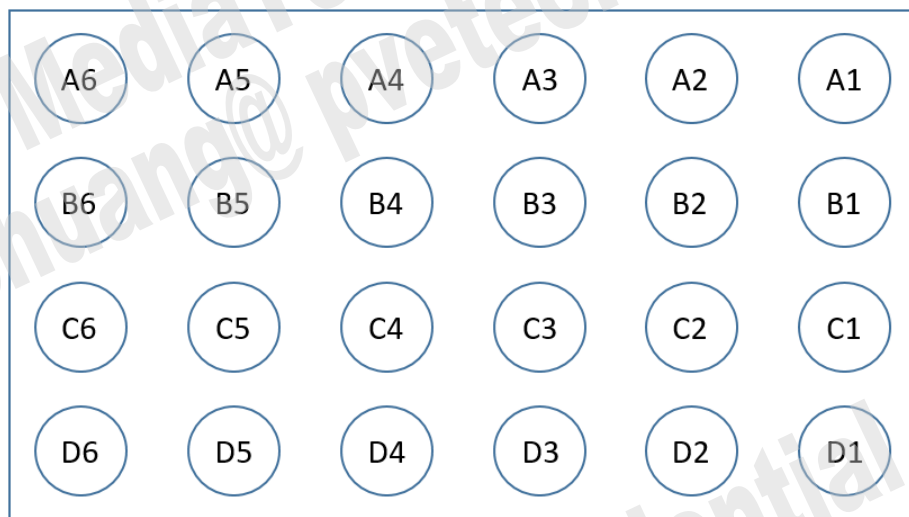


Figure 3-1. Overview of MT6686 pin map

3.2 Pin Descriptions

Table 3-1. MT6686 pin descriptions

Pin Name	Pin Location	I/O Type	Description	X	Y
Power Supply I/O					
AVDD_GNSS	A-3	P	1.6 to 1.8V supply	1324.78	1324.78
AVDD_XO	B-5	P	1.6 to 1.8V supply	546.78	935.78
DVDD_TOP	B-4	P	1.6 to 1.8V supply	935.78	935.78
DVDD13_CEXT	B-3	P	1.3V supply (for sleep mode only - optional)	1324.78	935.78
DVDD18_12_IO	D-6	P	1.8V/1.2V GPIO supply	157.78	157.78
Ground I/O					
AVSS_GNSS	A-2	GND	GNSS ground	1713.78	1324.78
AVSS_GNSS	B-2	GND	GNSS ground	1713.78	935.78
AVSS_XO	A-5	GND	XO buffer ground	546.78	1324.78
AVSS_XO	B-6	GND	XO buffer ground	157.78	935.78
DVSS	D-3	GND	Digital ground	1324.78	157.78
RX I/O					
GNSS_RF_IN_L1	B-1	AI	L1-band GNSS RF input	2102.78	935.78
GNSS_RF_IN_L5	A-1	AI	L5-band GNSS RF input	2102.78	1324.78
Baseband Analog Interface I/O					
GNSS_Q_L1	C-1	AO	L1-band GNSS Intermediate Frequency (IF) receiver IQ signals	2102.78	546.78
GNSS_I_L1	D-1	AO	L1-band GNSS IF receiver IQ signals	2102.78	157.78
GNSS_Q_L5	C-2	AO	L5-band GNSS IF receiver IQ signals	1713.78	546.78
GNSS_I_L5	D-2	AO	L5-band GNSS IF receiver IQ signals	1713.78	157.78
Digital I/O					
TOP_DATA	C-5	DIO	2-wire data signal	546.78	546.78
TOP_CLK	C-6	DI	2-wire clock signal	157.78	546.78
GPIO0	D-5	DIO	eLNA control from companion modem via SPI	546.78	157.78
GPIO1	D-4	DIO	eLNA control from companion modem via SPI	935.78	157.78
HRST_B	C-4	DI	Hardware reset from companion modem	935.78	546.78
Xtal I/O					
XO_IN	A-6	AI	Reference clock input (26 MHz/52 MHz)	157.78	1324.78

Table 3-1. MT6686 pin descriptions (continued)

Pin Name	Pin Location	I/O Type	Description	X	Y
Others					
SCAN_EN	A-4	DI	Tied to GND during operation. For testing only.	935.78	1324.78
DVDD28_FS	C-3	AIO	eFuse FSOURCE pin tied to GND during operation.	1324.78	546.78

4 Specifications

4.1 Electrical Characteristics

4.1.1 Absolute Maximum Ratings

Operating MT6686 IC beyond these specifications can damage the device. Functional operation is not guaranteed under these conditions. Functional operation under any absolute maximum condition for extended periods can affect the device reliability.

Table 4-1. Absolute maximum ratings

Pin #	Name	Parameter	Rating	Unit
B-3	DVDD13_CEXT	1.3V supply	1.48	V
B-4	DVDD_TOP	1.6 to 1.8V supply	1.98	V
A-3	AVDD_GNSS	1.6 to 1.8V supply	1.98	V
B-5	AVDD_XO	1.6 to 1.8V supply	1.98	V
D-6	DVDD18_12_IO	1.8V/1.2V supply	2.0	V
-	T _j	Max. junction temperature	125	°C
All pins		ESD protection (HBM)	±2,000	V
All pins		ESD protection (CDM)	±500	V

4.1.2 Operating Descriptions

Table 4-2. Recommended operating range

Pin #	Name	Parameter	Min.	Typ.	Max.	Unit
B-3	DVDD13_CEXT	1.3V supply	1.25	1.3	1.4	V
B-4	DVDD_TOP	1.6 to 1.8V supply	1.5	1.6/1.8	1.89	V
A-3	AVDD_GNSS	1.6 to 1.8V supply	1.5	1.6/1.8	1.89	V
B-5	AVDD_XO	1.6 to 1.8V supply	1.5	1.6/1.8	1.89	V
D-6	DVDD18_12_IO	1.8V/1.2V supply	1.71	1.8	1.89	V
-	T _j	Commercial junction operating temperature	0	25	115	°C
		Industry junction operating temperature	-20	25	125	°C
-	T _a	Operation temperature	-30	25	85	°C
-	T _{stg}	Storage temperature ⁽¹⁾	-40		125	°C

(1) For complete storage conditions, please refer to Section 6.2 Storage.

4.2 Supply Specifications

The following tables list the power supply requirements.

Table 4-3. DVDD13_CEXT specifications

Test Item	Min.	Typ.	Max.	Unit	Note
Output voltage, VCN13	1.25	1.3	1.4	V	Optional, for sleep mode only
Output current	-	-	0.1	mA	Static leakage current

Table 4-4. AVDD_GNSS/AVDD_XO/DVDD_TOP specifications

Test Item	Min.	Typ.	Max.	Unit	Note
Output voltage, VCN18	1.5	1.6 to 1.8	1.89	V	-
Output current	-	25	-	mA	L1 + L5

Table 4-5. DVDD18_12_IO specifications

Test Item	Min	Typ.	Max.	Unit	Note
Output voltage, VIO	1.71 1.14	1.8 1.2	1.89 1.26	V	-
Output current	-	-	0.1	mA	Static leakage current

4.3 Power On/Off Sequence

MT6686 uses two supply voltages, VIO and VCN18. The RSTB signal should only flag high 50 μ s after both VIO and VCN18 supplies are ready. The figure below shows the chip's power-on sequence.

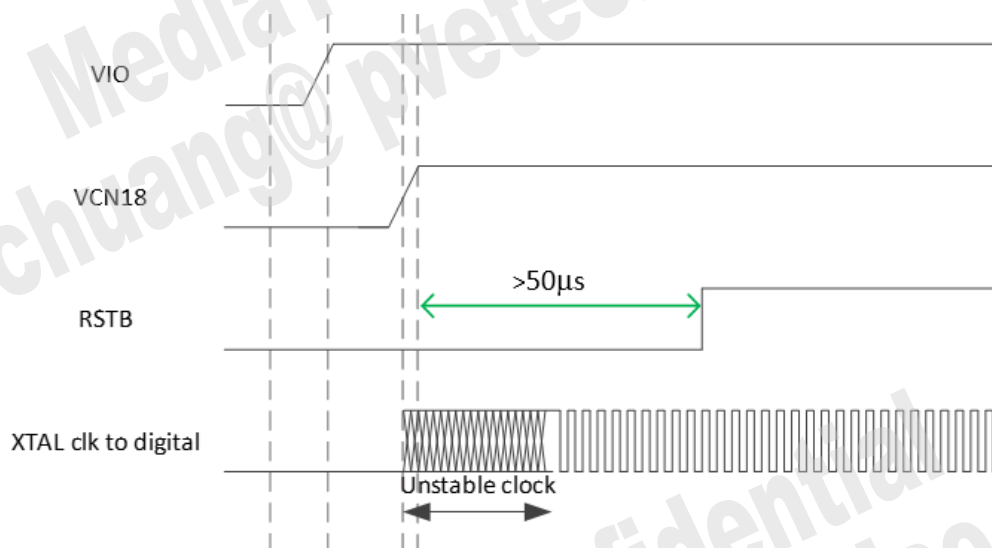


Figure 4-1. Power-on sequence

DVDD18 12 IO and AVDD_GNSS/AVDD_XO/DVDD_TOP

The I/O and internal control logic use the DVDD18_IO and AVDD_GNSS/AVDD_XO/DVDD_TOP supply pins. The supply voltages to these pins should be powered on in the sequence shown above for the logic to function properly. Inappropriate power-on sequence of DVDD18_IO and AVDD_GNSS/AVDD_XO/DVDD_TOP might lead to unexpected malfunction of MT6686.

DVDD13 CEXT

If sleep mode is used to cut off the current consumption on the AVDD_GNSS/AVDD_XO/DVDD_TOP pins, an external 1.3V supply is needed to keep the internal logic in the standby state. This supply voltage must be ready before entering the sleep mode.

4.4 Digital Logic Characteristics

MT6686 timing characteristics and interface protocols are introduced in the following sections including general comments.

4.4.1 Timing Diagram Convention

Figure 4-2 shows the conventions used with timing diagram throughout this document.

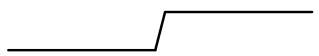
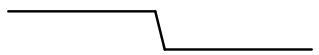
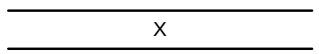
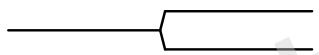
Waveform	Description
	Signal is changing from low to high
	Signal is changing from high to low
	Don't care or bus is driven
	Bus is changing from invalid to valid
	Bus is changing from high-Z to valid
	Denotes multiple clock periods

Figure 4-2. Timing diagram conventions

4.4.2 Rising/Falling Time Definition

Figure 4-3 is the rising and falling timing diagram. The actual signal timing curve is related to the external load conditions. See Table 4-6 for the operating conditions of digital logics.

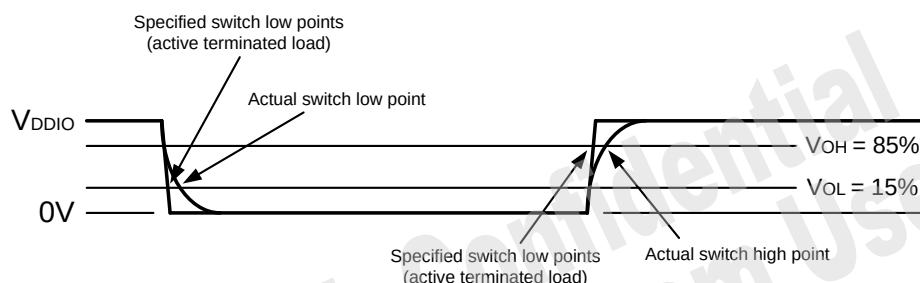


Figure 4-3. Rising and falling times diagram

Table 4-6. Operating conditions of digital logics

Parameter	Min.	Typ.	Max.	Unit	Note
VDD, supply of core power	1.17	1.3	1.43	V	-
VDDIO, supply of I/O power	1.62 1.08	1.8 1.2	1.98 1.32	V	-
VIH, input logic high voltage	0.7*VDD	-	VDD + 0.5	V	-
VIL, input logic low voltage	-	-	0.3*VDDIO	V	-
VOH (DC), DC output high voltage	0.7*VDD	-	VDD + 0.5	V	VDD = min, I _{OH} = 1.5 mA
VOL (DC), DC output low voltage	-	-	0.3*VDD	V	VDD = min, I _{OL} = 1.5 mA

4.4.3 Protocol

MT6686 uses the 2-wire control interface for the companion modem to program and control the chip.

4.4.3.1 2-Wire

The 2-wire bus of MT6686 is used by the SPI interface to access GPS and TOP command registers and configures/uses the GPIO0 and GPIO1 pins as external LNA enable control.

The bit number of SDATA depends on different operating conditions, as shown in Figure 4-4.

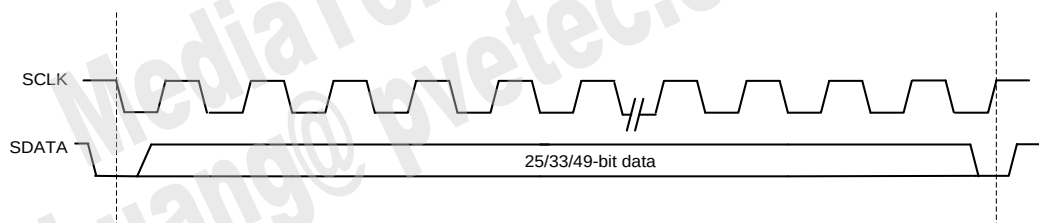


Figure 4-4. 2-wire SPI timing diagram

4.5 Reference Clock

The MT6686 device supports a clock of either 26 MHz (default) or 52 MHz. If the 52 MHz clock is used, initialize the DATA pin to high state before releasing HRSTB from low to high state upon power-up. See [Table 4-7](#) for the reference clock requirement.

Table 4-7. Reference clock requirement

Parameter	Condition	Min.	Typ.	Max.	Unit
Output frequency	-	-	26/52	-	MHz
Input loading	-	-	3	5	pF
Duty cycle	-	47	50	53	%
Rise/Fall time@20 to 80%	-	-	-	2	ns
Voltage swing	-	0.8	-	1.8	V
Phase noise	1 kHz	-	-127	-	dBc/Hz
	10 kHz	-	-139	-	dBc/Hz
	100 kHz	-	-148	-	dBc/Hz

[Table 4-8](#) shows MT6686 XO_IN input impedance.

Table 4-8. Pin (XO_IN) input impedance

Parameter	Condition	Min.	Typ.	Max.	Unit
XO_IN (Pin#A-6)	Input loading	-	3	5	pF

4.6 GNSS

4.6.1 GNSS Radio Description

The GNSS section consists of two separate RF receivers to provide simultaneous support of L1 and L5 frequency bands. For optimum co-existence performance, two external RF Surface Acoustic Wave (SAW) filters and a LNA must be placed before each of the receiver (see Figure 1-1).

For each RF receiver (L1 or L5 frequency bands), the simultaneous reception of signals from one or multiple constellations is made possible by configuring the local oscillator and RX filter for different GNSS modes.

4.6.2 GNSS Mode Definition

L1 band operation

- GPS-only mode or Galileo-only mode
- GPS + Galileo mode
- GPS + GLONASS, Galileo + GLONASS, or GPS + Galileo + GLONASS mode
- GPS + BeiDou, Galileo + BeiDou, or GPS + Galileo + BeiDou mode
- GPS + GLONASS + BeiDou, Galileo + GLONASS + BeiDou, or GPS + Galileo + GLONASS + Galileo mode

The receiver sets LO to 1582.464 MHz and intermediate frequency filter to LPF with bandwidth of 24 MHz.

L5 band operation

- E5a mode
- E5a/E5b mode

Note: E5b is not supported by MT2737.

4.6.3 GNSS Specifications

Note:

- (1) The specification value is valid at room temperature (25°C).
- (2) All specifications are measured at the RF port unless otherwise specified.
- (3) The system performance depends on the companion modem chip's capability.

Table 4-9. GNSS RF specifications (GPS-only mode or Galileo-only mode or GPS/Galileo mode)

Parameter	Condition	Min.	Typ.	Max.	Unit
RF input frequency	-	-	1575.42	-	MHz
Maximum input level	Peak power of close-in out-of-band jammer (beyond the input frequency specified above) at RFIC input	-	-	-17	dBm
Input return loss	Single-ended input and externally matched to 50Ω source using matching network for all gain	-10	-	-	dB
Gain (Av)	Low current mode with max. PGA gain	-	58	-	dB
NF	High current mode with max. PGA gain	-	3	-	dB

Table 4-9. GNSS RF specifications (GPS-only mode or Galileo-only mode or GPS/Galileo mode) (continued)

Parameter	Condition	Min.	Typ.	Max.	Unit
Image rejection	All mode	-	35	-	dB
DC offset	-	-	±50	±100	mV
RX current	High current mode	-	7.7	-	mA

Table 4-10. GNSS RF specifications (GPS/GLONASS mode or Galileo/GLONASS mode)

Parameter	Condition	Min.	Typ.	Max.	Unit
RF input frequency	GPS/GLONASS	-	1575.42/ 1601.71	-	MHz
Maximum input level	Peak power of close-in out-of-band jammer (beyond the input frequency specified above) at RFIC input	-	-	-17	dBm
Input return loss	Single-ended input and externally matched to 50Ω source using matching network for all gain	-10	-	-	dB
Gain (Av) (integrated average over Fc ±4M)	Low current mode with max. PGA gain	-	58	-	dB
NF (integrated average over Fc ±4M)	High current mode with max. PGA gain	-	3	-	dB
Image rejection	All gain mode	-	35	-	dB
DC offset	-	-	±50	±100	mV
RX current	High current mode	-	8.5	-	mA

Table 4-11. GNSS RF specifications (GPS/BeiDou mode)

Parameter	Condition	Min.	Typ.	Max.	Unit
RF input frequency	GPS/BeiDou	-	1575.42/ 1561	-	MHz
Maximum input level	Peak power of close-in out-of-band jammer (beyond the input frequency specified above) at RFIC input	-	-	-17	dBm
Input return loss	Single-ended I input and externally matched to 50Ω source using matching network for all gain	-10	-	-	dB
Gain (Av) (integrated average over Fc ±4M)	Low current mode with max. PGA gain	-	58	-	dB
NF (integrated average over Fc ±2M)	High current mode with max. PGA gain	-	3	-	dB
Image rejection	All gain mode	-	35	-	dB
DC offset	-	-	±50	±100	mV
RX current	High current mode	-	8.5	-	mA

Table 4-12. GNSS RF specifications (GPS/GLONASS/BeiDou mode or GPS/GLONASS/Galileo mode)

Parameter	Condition	Min.	Typ.	Max.	Unit
RF input frequency	GPS/Galileo/GLONASS/BeiDou	-	1575.42/ 1601.71/ 1561	-	MHz
Maximum input level	Peak power of close-in out-of-band jammer (beyond the input frequency specified above) at RFIC input	-	-	-17	dBm
Input return loss	Single-ended input and externally matched to 50Ω source using matching network for all gain	-10	-	-	dB
Gain (Av) (integrated average over Fc ±4M)	Low current mode with max. PGA gain	-	58	-	dB
NF (integrated average over Fc ±2M)	High current mode with max. PGA gain	-	3	-	dB
Image rejection	All gain mode	-	35	-	dB
DC offset	-	-	±50	±100	mV
RX current	High current mode	-	10.6	-	mA

Table 4-13. GNSS RF specifications (GPS-L5/Galileo E5a/IRNSS/BeiDou-3 B2a mode)

Parameter	Condition	Min.	Typ.	Max.	Unit
RF input frequency	GPS/GLONASS	-	1176.45	-	MHz
Maximum input level	Peak power of close-in out-of-band jammer (beyond the input frequency specified above) at RFIC input	-	-	-17	dBm
Input return loss	Single-ended input and externally matched to 50Ω source using matching network for all gain	-10	-	-	dB
Gain (Av) (integrated average over Fc ±4M)	Low current mode with max. PGA gain	-	58	-	dB
NF (integrated average over Fc ±4M)	High current mode with max. PGA gain	-	3	-	dB
Image rejection	All gain mode	-	35	-	dB
DC offset	-	-	±50	±100	mV
RX current	High current mode	-	10.5	-	mA

5 Part Reliability

5.1 Qualification Summary

Table 5-1. MT6686 product reliability qualification

Test Item and Condition	Sample Size	Result	Reference
HTOL Tj = 125°C Vccmax for core and I/O	77 units x 3 lots	Pass	JESD22-A108
ESD - Human body mode 25 ± 5°C, 55 ± 10% R.H. (±)2000V	Each condition x 3 units	Pass	JS-001-2017
ESD - Charged device mode 25 ± 5°C, 55 ± 10% R.H. 500V(+), 500V(-)	Each condition x 3 units	Pass	JS-002-2018
Latch up (I-test) Class II 100mA	Each condition x 3 units	Pass	JEDEC EIA/JESD78
Latch up (Overvoltage) Class II 1.5 * Vccmax	Each condition x 3 units	Pass	JEDEC EIA/JESD78

Table 5-2. MT6686 package reliability qualification

Test Item and Condition	Number of Assembly Lots	Sample Size per Lot	Result (Judged By Functional Test and SAT)	Reference
Preconditioning MSL1	3	75	Pass	J-STD-020
Temperature cycling Condition B (-55°C to +125°C) 1,000 cycles	3	25	Pass	JESD22-A104
Unbiased HAST 130°C / 85% R.H., 230 Kpa 96 hrs	3	25	Pass	JESD22-A118
Biased HAST 110°C/85% R.H., 264 hrs Vcc max for core & IO	3	25	Pass	JESD22-A110
High temperature storage test (HTST) 150°C, 1,000 hrs	3	25	Pass	JESD22-A103

5.2 Device Characteristics

Table 5-3. MT6686 device characteristics

Device Name	MT6686
Package Type	WLCSP
Package Body Size	2.17x1.42x0.5mm
Lead Composition	SAC405
Fab Process	55 nm
Fab Site	TSMC
Assembly Site	Advanced Semiconductor Engineering, Inc (ASE), Taiwan Siliconware Precision Industries Co., Ltd. (SPIL), Taiwan
Solder Ball Pitch	0.35mm

6 Packing Information

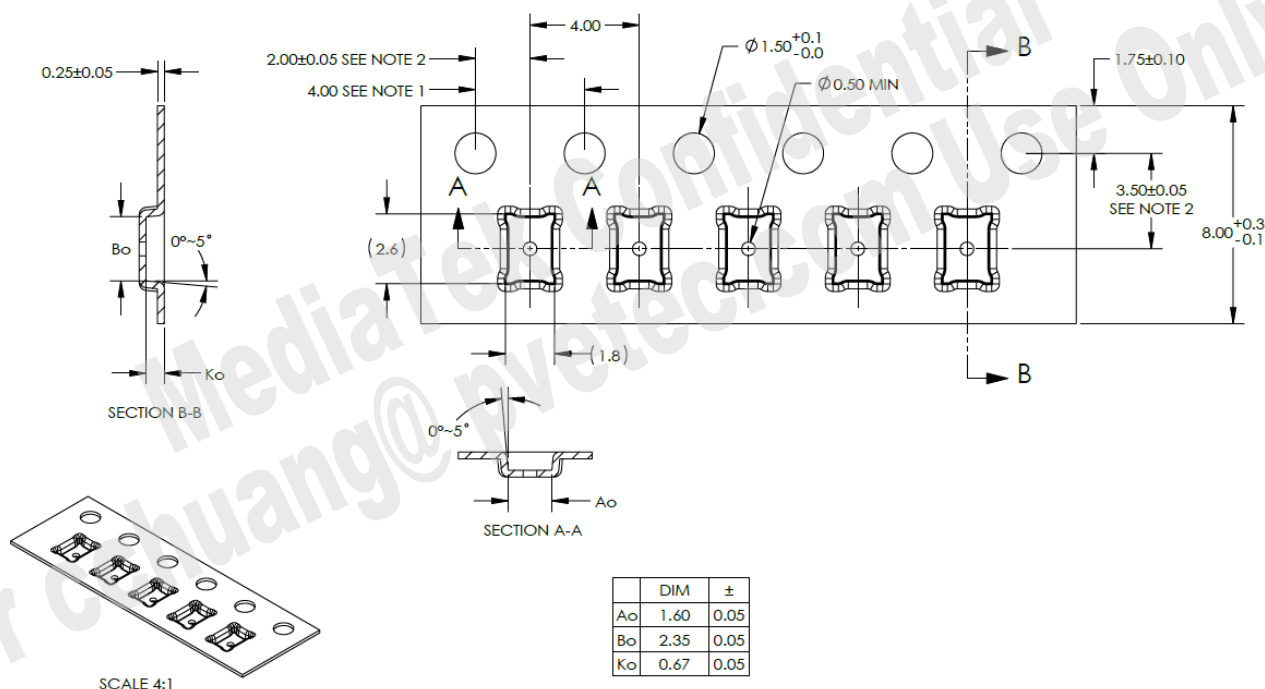
Package	Size (mm)	EA/Roll	Tape Reel/Cartron	Full Carton Q'ty
WLCSP	2.1655 x 1.4203	3,000	5	15,000

6.1 Carrier

6.1.1 Tape and Reel Information

The tape carries system conforms to EIA-541 standards.

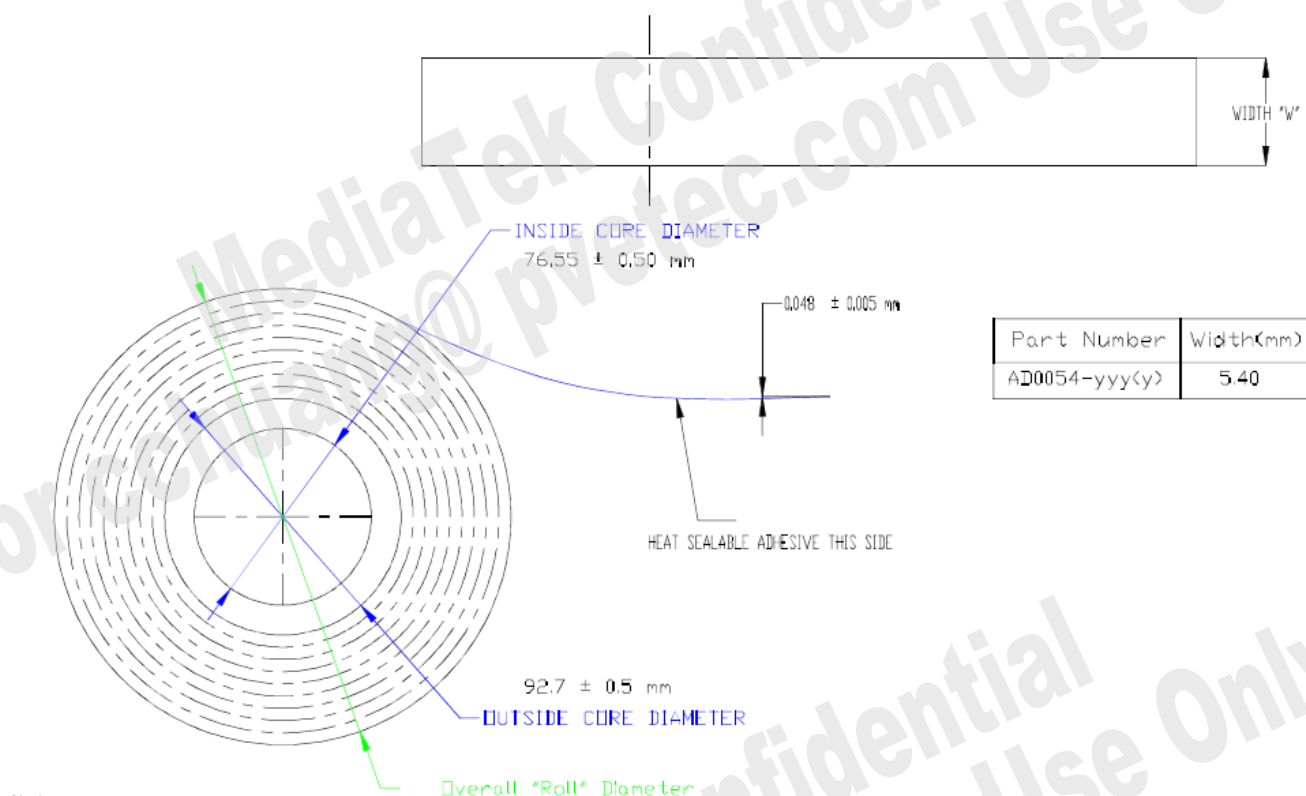
An illustration of the MT6686 tape carrier and the corresponding dimensions are shown below of 3,000 EA per roll.



NOTES:

1. 10 SPROCKET HOLE PITCH CUMULATIVE TOLERANCE ± 0.2
2. POCKET POSITION RELATIVE TO SPROCKET HOLE MEASURED AS TRUE POSITION OF POCKET, NOT POCKET HOLE.
3. Ao AND Bo ARE MEASURED ON A PLANE AT A DISTANCE "R" ABOVE THE BOTTOM OF THE POCKET.

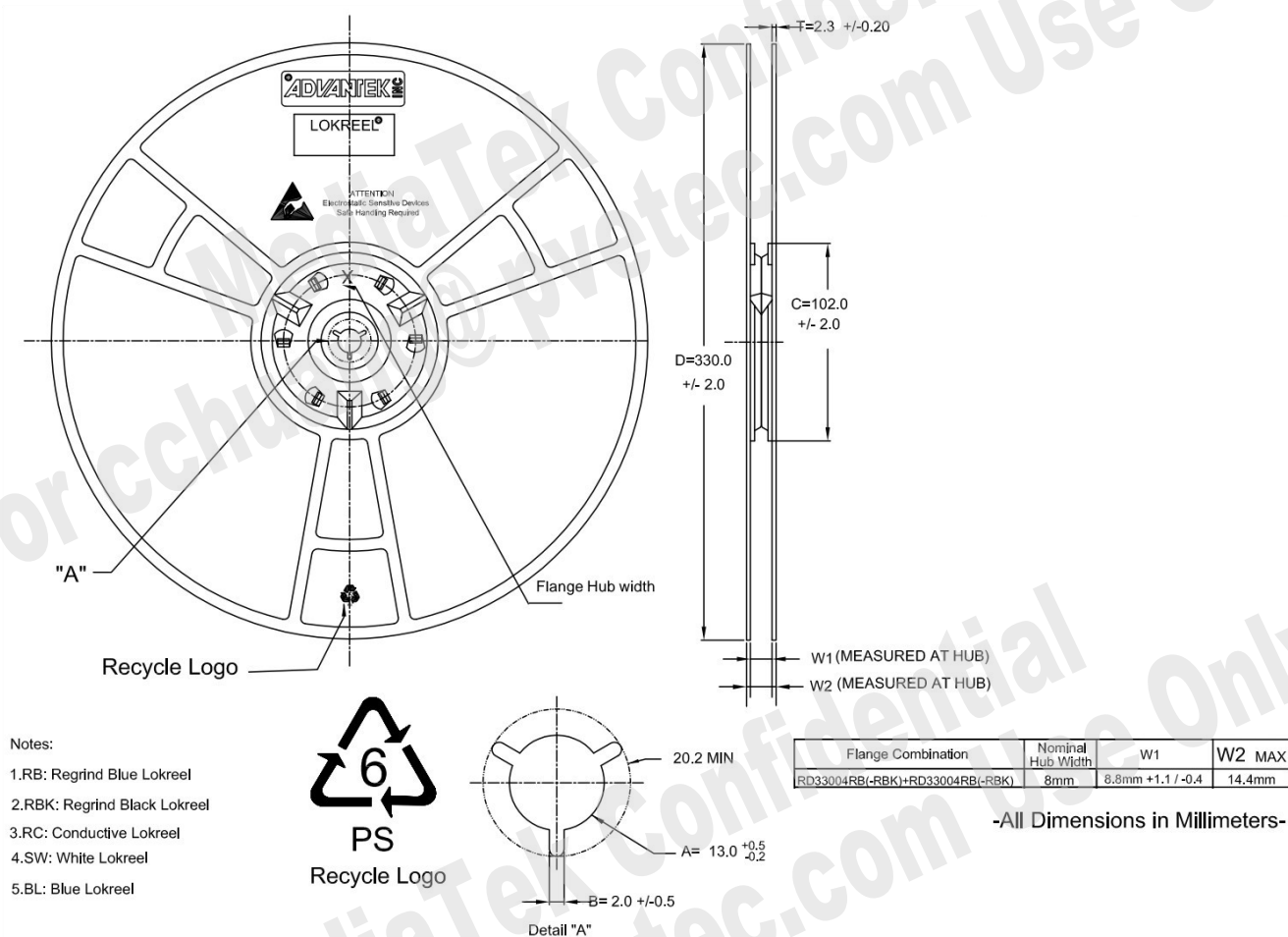
Figure 6-1. Carrier tape dimensions



Note:

1. Tape Width tolerance ± 0.10 mm
2. Length (M) tolerance:
 For 300M length type = $+5.0 / - 0.0$ M
 For 490M length type roll = $+5.0 / - 0.0$ M
 For 500M length type roll = $+1.0 / - 0.0$ M
3. Please refer to Tech Bulletins for related properties of cover tape
4. Overall 'Roll' Diameter:
 For 300M length type roll = $163 +1.0 / -0.0$ mm
 For 490M length type roll = $196 +1.0 / -0.0$ mm
 For 500M length type roll = $197 +1.0 / -0.0$ mm
5. Core width dimension refer to below table

Figure 6-2. Cover tape



6.1.2 Aluminum Foil Bag

Table 6-1. Aluminum foil bag specification

Size	Thickness	Surface Impedance
400 x 430 mm	0.15 x 0.005 mm	$10^5 - 10^{12}$ Ohm/SQ

6.1.3 Box and Carton

	Material	Strength	Size	Printing
Box	3 Layer-B corrugated paper	12 Kgf/cm	340(L) x 325(W) x 50(H) mm	Black (script, warning symbol, indicate symbol RoHS)
Carton	5 Layer-AB corrugated paper	18 Kgf/cm	374(L) x 285(W) x 363(H) mm	Black (script, warning symbol, indicate symbol RoHS)

6.2 Storage

6.2.1 Bagged

The MT6686 devices must be delivered in sealed and aluminum foil bags (refer to [Table 6-1](#) for the specification) along with desiccants of 110 x 120 mm weighing 66g. The components should be packaged within tape and reel carriers, adhering to warehouse control procedures utilizing the FIFO method. The shelf life is unlimited and does not require special storage conditions.

6.2.2 Out-of-Bag

The device does not require special storage conditions provided, and it is maintained at conditions equal to or less than 30°C /85%RH.

7 Package Information

7.1 Top Marking

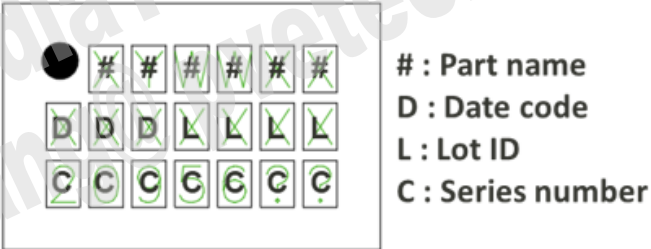


Figure 7-1. Top marking of MT6686

7.2 Ordering Information

Table 7-1. MT6686 ordering information

Order Number	Marking	Operational Temperature Range	Package
MT6686P/A	MT6686	-40 to 85°C	WLCSP



ESD CAUTION

MT6686 is an ESD (electrostatic discharge) sensitive device and may be damaged with ESD or spike voltage. Although MT6686 comes with built-in ESD protection circuitry, please handle it with care to avoid the permanent malfunction or the performance degradation.

7.3 Mechanical Drawing

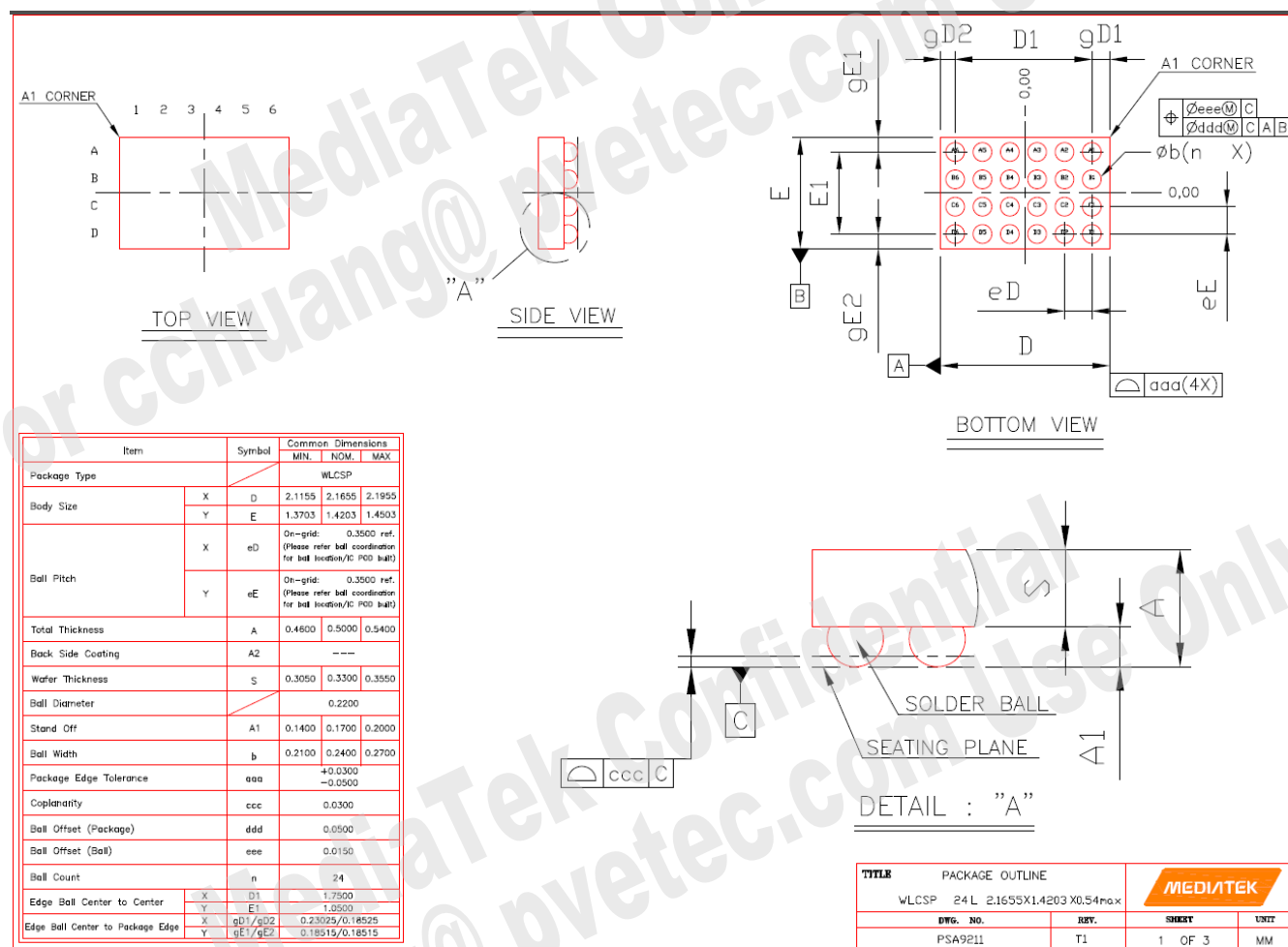


Figure 7-2. Physical dimension of MT6686

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